

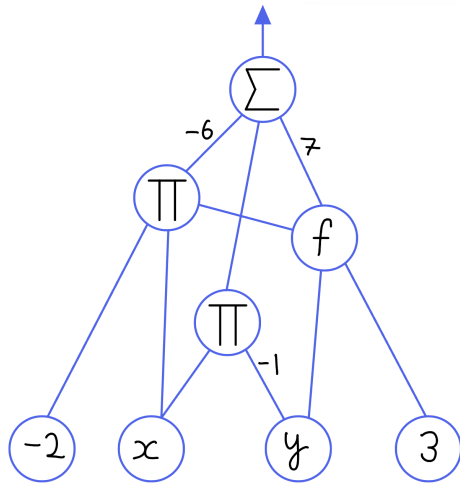
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Debordering Closure Results in Determinantal and Pfaffian Ideals

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Algebraic Circuits with Oracles



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

Figure: $-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the *circuit complexity* of some family of polynomials.

Definition

Fix some polynomials $g_1(\bar{x}), \dots, g_k(\bar{x}) \in \mathbb{F}[\bar{x}]$.

The *ideal generated by* $g_1(\bar{x}), \dots, g_k(\bar{x})$ is the set of combinations

$$\langle g_1, \dots, g_k \rangle = \left\{ \sum_{i=1}^k h_i(\bar{x}) \cdot g_i(\bar{x}) \mid h_i(\bar{x}) \in \mathbb{F}[\bar{x}] \right\}.$$

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

Principal Ideals

Example

The *principal ideals* are generated by a single polynomial g .
If $f \in \langle g \rangle$, then g is a *factor* of f .

Question

Suppose $f \in \langle g \rangle$. Does g have a small f -oracle circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If g is a factor of f , $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *border computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

By border computation, we mean the circuit computes the following:

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}(\varepsilon)[\overline{x}], \quad q \geq 1.$$

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = n^{\Theta(1)}$.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(\overline{x})$ in a standard basis indexed by combinatorial objects, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Homogenization* as well as the *Isolation Lemma*.

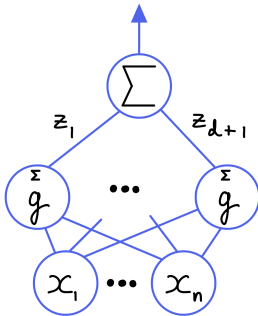
Homogenization

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.



Then, we can compute $\text{coeff}_{t^i}(g)$ by a size $O(ds)$, depth Δ f -oracle circuit.

Issues with Homogenization

If a circuit border computes $g(\overline{x})$, then it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon)$$

for some $q \geq 1$.

Idea: Homogenize with respect to ε .

Problem: q can be *arbitrarily large*

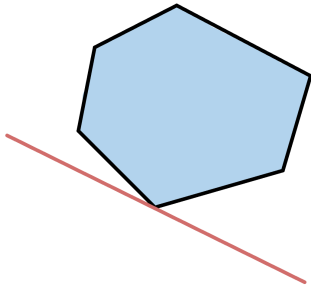
$\implies$ Homogenization gives *large circuit*.

Isolation Lemma

In our proof, we have a specific monomial in $f(\bar{x})$ we want to *isolate*.

Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.



Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be choose to be small
 $\implies$ smaller blowup in circuit size $\implies$ homogenization w.r.t w is small.

Thank You!

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